ABUSE.CH URLHAUS API TEST SUITE

Malware Payload Distribution & Community Threat Feed Verification Benchmark

API Integration Under Test: Abuse.ch URLhaus API v1 (https://urlhaus-api.abuse.ch/v1/url/)

Configuration: Enter your key in Settings > API Keys > 'URLhaus API Key' or set URLHAUS_API_KEY in .env

FileSanitizer posts queries to the URLhaus database, surfacing threat tags, malware families, and reporters.

USECASE 1: KNOWN MALWARE DISTRIBUTION PAYLOAD LINK

MALICIOUS

URL Target: http://evil.com/payload.exe

Active executable payload URL listed in malware databases. Exercises URLhaus API blacklist checking.

Expected Result: reputation.urlhaus = 'MALICIOUS', risk = 'MALICIOUS', threat tag indicator.

URLhaus Reference: https://urlhaus.abuse.ch/browse/

USECASE 2: DIRECT SUSPICIOUS IP WITH EXECUTABLE EXTENSION

HIGH RISK

URL Target: http://198.51.100.22/malware.exe

Demonstrates hybrid heuristic analysis: direct public IP host address + dangerous .exe extension + URLhaus lookup.

Expected Result: Heuristic flag (Dangerous extension) + URLhaus cross-reference check.

URLhaus Reference: https://urlhaus.abuse.ch/api/

USECASE 3: CLEAN DOMAIN NOT IN URLHAUS DATABASE (BENIGN)

CLEAN

URL Target: https://urlhaus.abuse.ch

Official Abuse.ch URLhaus project portal. Query status returns 'no_results', validating clean parsing.

Expected Result: reputation.urlhaus = 'CLEAN', zero malicious records reported.

URLhaus Reference: https://urlhaus.abuse.ch

USECASE 4: PYTHON SOFTWARE FOUNDATION REPOSITORY

CLEAN

URL Target: https://www.python.org

Legitimate programming language distribution portal validating high-reputation domain handling.

Expected Result: reputation.urlhaus = 'CLEAN', TLS certificate valid, screenshot captured.

URLhaus Reference: https://www.python.org

USECASE 5: THREAT INTEL TOGGLE IN SETTINGS

Admins can disable external threat intel lookups via Settings > External Connections > 'Allow Threat Intel Lookups'.

When disabled, URLhaus and VirusTotal external API connections are suppressed, enforcing an isolated zero-touch audit.

Internal heuristic scanners (executable extensions, raw IP checks, keywords) continue functioning autonomously.